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Working with Data in Excel

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Open Scholarship and Data Services

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Topics Covered

- Functions
 - Basic math, dates, counting cells, manipulating text, if-then statements
- Filtering
- ~~• Conditional formatting~~
- PivotTables

Data

- <https://github.com/keikob/Workshops>
 - Bookstore_Workshop_Dataset.xlsx

Functions

Basic mathematical functions

- SUM()
 - Adds together the numbers in a range
- AVERAGE()
 - Finds the average of a range of numbers
- STDDEV.S()
 - Finds the standard deviation of a range of numbers (treating them as a sample of a population)

Logical tests

- Equal to =
- Greater than >
- Less than <
- Greater than or equal to >=
- Less than or equal to <=
- Not equal to <>

Boolean operators

- AND()
 - Unlimited list of arguments, all containing logical tests (e.g., >5) that will be evaluated as TRUE or FALSE
 - Returns TRUE if all statements are true, and FALSE if any are false
- OR()
 - Unlimited list of arguments, all containing logical tests (e.g., >5) that will be evaluated as TRUE or FALSE
 - Returns TRUE if any statements are true, and FALSE if all are false
- NOT()
 - Single argument, containing a logical tests
 - Returns the opposite of the result (e.g. FALSE for TRUE)

Date functions

- YEAR, MONTH, DAY
 - Extracts the year, month, or day portion of a date
 - =YEAR(date), =MONTH(date), =DAY(date)
 - Ex. =YEAR(10/9/26) → 2026, =MONTH(10/9/26) → 10, =DAY(10/9/26) → 9
- TEXT
 - Converts a date to text in a specific number format
 - =TEXT(10/9/26, "mmm") → Oct
 - =TEXT(10/9/26, "mmmm") → October
 - =TEXT(10/9/26, "ddd") → Fri
 - =TEXT(10/9/26, "dddd") → Friday

Formulas that count cells

- COUNT
 - Counts cells that contain numbers
 - =COUNT(range)
 - COUNT(A1:A20)
- COUNTA
 - Counts cells that contain text
 - =COUNTA(range)
 - Ex. =COUNTA(A1:A20)
- COUNTBLANK
 - Counts cells that do not contain text
 - =COUNTBLANK(range)
 - Ex. =COUNTBLANK(A1:A20)

Formulas that count cells

- COUNTIF
 - Counts cells that meet a specific condition
 - =COUNTIF(range, criteria)
 - Ex. =COUNTIF(A1:A20, ">5"), =COUNTIF(A1:A20, "abc")
- COUNTIFS
 - Counts cells that meet multiple conditions
 - =COUNTIFS(range1, criteria1, range2, criteria2)
 - Ex. =COUNTIFS(A1:A20, ">5", B1:B20, "abc")

Manipulating text

- LEFT() and RIGHT()
 - Returns the specific number of characters from the left/right of a cell
 - =LEFT(cell, num_of_characters)
 - Ex. =LEFT(A1, 3)
- LEN()
 - Returns the number of characters in a cell
 - = LEN(cell)
 - Ex. =LEN(A1)
- &
 - Combines text from multiple cells
 - = cell & cell
 - Ex. =A1&B1
 - To include your own text, =A1&"text"

If-then statements

- IF
 - Checks whether a condition is met, and returns one value if TRUE, and another if FALSE
 - =IF(logical_test, value_if_true, value_if_false)
 - Ex. =IF(A1>5, "Yes", "No")
- IFS
 - Checks whether one or more conditions are met, and returns a value corresponding to the first TRUE condition
 - =IFS(logical_test1, value_if_true1, logical_test2, value_if_true2, ...)
 - Ex. =IFS(A1>500, "large", A1>300, "medium", A1>100, "small", A1<=100, "tiny")

Filters

Adding filters

- On the Home tab, click on the Sort & Filter button.
 - Depending on the size of your screen, this may appear as an icon with A, Z, and a funnel.
- Click on Filter to turn on filters.
- Drop-down arrows will appear next to each column header. When you click on one, it will give you a list of all the different values that appear in that column, with checkboxes beside them.
- Check the boxes next to the values you want to look at.
- The spreadsheet will update to only show the rows that match that value.

Filter functionality

- Alternative to COUNT functions
 - After applying a filter, the number of cells left in the display will be shown in the bottom left of the screen (ex. "50 of 200 records found")
- Alternative to IF functions
 - To create a categorical variable based on an existing column, filter the data and fill in only the visible cells, then refilter and fill in the remaining cells.
 - Ex.
 1. Filter column A to values >50, then fill in the cells of column B with "large".
 2. Reset column A filter to Select All, then filter column B to just the blank cells, then fill in the cells of column B with "small".

More filter info

- You can filter using multiple columns at a time, to find the intersection of the two conditions (similar to AND).
- You can tell when a filter is applied because the row numbers on the right will be blue.
- Always make sure that you have reset all filters back to “Select all” before making new calculations that you want to apply to the whole dataset!

PivotTables

What is a pivot table?

- When you have data with a lot of variables, you often want to be able to summarize it in multiple different ways.
- You may want to look at the dependent variable(s) across several different predictors, and you may want to combine these predictors together or nest them inside each other.
- Excel's PivotTable function is a way to have Excel summarize your data in a table format for you, and to quickly shuffle categories around to look at the data in different ways.

How to insert a pivot table

1. Go to the Insert tab.
2. With a cell inside your data selected, click PivotTable.
3. A pop-up will appear asking for your range of data. It should have automatically selected your full data frame, but you can check to make sure all the columns and rows are included.
4. Tell it to send the pivot table to a new worksheet. (This should already be selected.)
5. This will pull up a new sheet with an empty area for a table, and a panel on the right containing a list of column headings.
6. Drag an independent variable to the box labeled Rows, and a dependent variable to the box labeled Values.

Types of values

- **Sum:** all of the values within a category are added together
 - **Count:** all the cells within a category are counted
 - **Average:** all of the values within a category are averaged
 - Other types: maximum, minimum, product, standard deviation...
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- Select the value type by clicking on the drop-down arrow under Values and choosing Value Field Settings.
 - The pivot table will automatically select Sum by default, but in most cases, you'll want to use Average.

Sorting and grouping

- To sort a column, such as by alphabetical order or from largest to smallest, right-click in the column you want, go to Sort, and choose "Sort smallest to largest" or "Sort largest to smallest".
- To manually shuffle rows or columns, click on a cell and drag it.
- To group rows together:
 - For non-numeric variables, highlight the labels you want to group together, right-click, and choose Group. Then do the same to any other labels you want to group together.
 - For numeric variables, right-click on a label in the leftmost column and click Group. Choose what you want your range to be, and how wide you want the bins to be. For example, to divide the years 1973-1999 into 1970s, 1980s, and 1990s, you would choose "Starting at: 1970", "Ending at: 1999", and "By: 10".

Multidimensional tables

- Including multiple variables in the Rows field means that your second variable will be listed as a subset of the first
- Including multiple variables in the Values field means that each dependent variable is listed as a separate column.
- Including variables in both Rows and Columns means that values are listed at the intersection of a column and row, rather than as a separate column.
 - You can only have one dependent variable when you do this!

Filters

- Filters allow you to get rid of data you don't want, or to focus on a subset of your data.
- Drag a variable into the Filters field.
- Click the drop-down arrow next to that variable above the pivot table and select the level(s) you want to observe.

Changing data

- If you change any part of your data, you can update the pivot table by clicking Refresh on the PivotTable Analyze tab.
- If you need to add or delete rows or columns, go to Change Data Source and edit the range.

PivotCharts

- To add a graph based on your pivot table:
- On the PivotTable Analyze tab, select PivotChart.
- This will bring up Excel's standard graph functions, and you can choose a chart type.
- Whenever you shuffle around the fields on your pivot table, the graph will change to match.
- Any labels that you change on the pivot table will also change on the graph.

Upcoming Workshops

- Data Wrangling in Python (Oct 16, 11am)
- Crafting Authentic Job Applications with AI Tools (Oct 16, 12:15pm)
- Becoming a Savvy AI User: What Everyone Should Know (Oct 29, 4pm)
- AI Tools in the Workplace (Nov 5, 4pm)
- Graphing Basics for R Users (Nov 6, 11am)

[Register here](#)

Are there any workshops you would like to see offered? [Let us know!](#)

Data Consultations

- <https://calendar.libs.uga.edu/appointments/osds>
- Make an appointment to sit down with someone from OSDS and talk about your research
 - In person or virtual
- Support for:
 - R, Python, Excel, GIS
 - Statistical analysis, text analysis, qualitative data analysis
 - Data visualization
 - Data management planning
- Or attend walk-in hours in the AI & Data Studio
 - Tuesdays 10-11 am
 - Thursdays 2-3 pm

Contact Us

<https://guides.libs.uga.edu/data-services/>

- Ask us a question
- Schedule a consultation
- Register for a workshop
- Find resources for learning about tools
- Visit the AI & Data Studio